

ECOAI BUILDING QUALIFICATION

Building Review & Qualification Guide

How we identify where continuous HVAC optimisation can deliver measurable value

Business case first. Technical confidence before implementation.

The review is designed to establish whether a building offers a credible energy, carbon and commercial opportunity before either party commits significant time or resource.

Why start with a building review?

Not every building offers the same opportunity. EcoAI reviews how the building operates, the available energy data and the existing BMS infrastructure so effort is focused where continuous HVAC optimisation is most likely to deliver meaningful, measurable energy savings.

The objective is not to find buildings that need replacing.

It is to identify buildings where the systems already in place can perform significantly better.

What the initial review considers

Commercial opportunity

Is there enough HVAC energy use and avoidable waste to create a meaningful energy and financial case?

Existing infrastructure

Can the BMS provide useful HVAC data and allow safe, controlled changes to selected setpoints or offsets? It does not need to be new or sophisticated.

Measurement

Is there sufficient meter and operating data to understand current performance and verify future energy savings?

Operational context

How is the building used, what comfort conditions matter, where does engineering effort go today, and are major changes already planned?

CANDIDATE BUILDING

What makes a strong candidate building?

We look for a combination of opportunity, controllability, evidence and a stable route to value.

Six characteristics matter most

1

Meaningful HVAC energy use

A sufficiently large heating, cooling or ventilation load to create a worthwhile energy-saving opportunity.

2

Existing BMS infrastructure

A live BMS with useful HVAC data and controllable setpoints or offsets. The existing BMS remains the master control system.

3

Operational opportunity

Changing schedules, weather response, occupancy, setpoint drift, out-of-hours operation, simultaneous heating/cooling or repeated manual tuning can create scope for continuous optimisation.

4

Good measurement potential

Suitable meter and operating data make it possible to establish a credible baseline and verify performance.

5

Reasonably healthy controls and plant

Sensors, valves, dampers, actuators and core control loops should be broadly functional. Optimisation cannot compensate for failed plant or fundamentally broken controls.

6

A stable operating future

The building is expected to remain in use and is not about to undergo a major change that would make the current assessment obsolete.

A strong candidate is not simply an old or inefficient-looking building.

It is a building with a credible energy and commercial case, useful BMS visibility, controllable HVAC opportunity and enough operational stability to make measurement meaningful.

INITIAL INFORMATION

What information is useful at the start?

Enough to understand the opportunity - not a full technical survey.

Approximate information is sufficient for the initial assessment.

EcoAI can start with the information available, identify any gaps and agree what is genuinely worth confirming next. The objective is to avoid creating unnecessary work before the opportunity is proven.

Useful initial information

- Building name, location, use and approximate floor area
- Annual electricity and heating consumption and, where available, energy cost or unit rates
- BMS make, platform/version and BMS/controls contractor if known
- Main HVAC systems - for example AHUs, heating circuits, cooling circuits and major plant
- Typical operating hours and occupancy patterns
- Known comfort, plant or controls issues
- Known repeated manual interventions, overrides or tuning requirements
- Planned major plant, controls or refurbishment works
- Available meter data, interval data and any existing energy reporting

Who normally helps provide it?

Estates / FM

Building use, schedules, comfort requirements, operational issues and planned works.

Energy / sustainability

Consumption data, tariffs, meter information and performance reporting.

BMS / controls

BMS platform, architecture, point information and likely read/write access route.

IT / OT

Connectivity and security requirements once a credible building opportunity is identified.

QUALIFICATION JOURNEY

From initial review to Technical Validation

A clear progression from opportunity to evidence.

The Building Review journey

1 Initial information Gather the available building, energy and BMS information.	2 Desktop assessment Review the indicative energy, carbon and commercial opportunity using the information available.	3 Client / engineering review Confirm assumptions, identify gaps and agree whether the opportunity merits further work.	4 Technical Validation Confirm BMS read/write capability, selected control points, control health, metering, comfort limits, secure connectivity and the integration route.
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Desktop assessment

The desktop stage answers a simple question: is there enough evidence of a meaningful opportunity to justify Technical Validation? It is indicative, based on available information, and is not a technical audit or formal offer.

Client / engineering review

Where the opportunity is credible, EcoAI works with the client and relevant engineering teams to confirm the building context, refine assumptions and establish what additional information is worth gathering.

Technical Validation

Technical Validation tests the proposed optimisation scope in detail before implementation. It confirms the viable control points, read/write route, core control health, metering, comfort requirements, operating constraints, secure connectivity and deployment route.

Technical Validation is the bridge between a credible business case and controlled implementation

It confirms the opportunity before either party commits significant implementation resource.

DECISION

What happens after the review?

A clear recommendation based on the evidence available.

Three possible outcomes

Proceed to Technical Validation

The energy and commercial opportunity appears strong and there is a credible route to deployment. The next step is to confirm the technical scope and implementation requirements.

Further information required

The opportunity may be attractive, but additional energy, BMS or operational information is needed before a confident progression decision can be made.

Not currently suitable

The building does not presently offer a strong enough opportunity, or planned changes mean it is more sensible to pause and revisit later.

If the building progresses

- Deployment is planned with the client, BMS/controls and IT/OT teams as required.
- myCoreAI is typically deployed remotely with minimal disruption to normal building operations.
- The existing BMS remains the master controller; existing plant safeties and local logic remain in place.
- Building data is collected so the system can learn how the building behaves.
- Selected setpoints and offsets are optimised within agreed comfort and operational limits.
- Performance is measured and energy savings are verified against the agreed baseline.

The purpose of the Building Review is simple:

Identify the right buildings, establish the energy and commercial case, and confirm there is a credible technical route before either party commits significant resource.

Business case first. Technical confidence before implementation.